

Does Housing Re-Price Disaster Risk Only Where People Believe? Exposure Difference-in-Differences Around Hurricanes Ian and Helene

Mateo Eduardo Stine

Lei Gao*

July 2026

Abstract

A prominent hypothesis in climate finance holds that housing markets capitalize disaster risk *only where residents believe* the risk is real (Baldauf et al., 2020). We test this after two salient landfalls—Hurricane Ian (SW Florida, Sept. 2022) and Hurricane Helene (Big Bend Florida to southern Appalachia, Sept. 2024)—using an exposure difference-in-differences on monthly Zillow county home values. Treated counties are those FEMA designated for its Individuals & Households Program (hard-hit); controls are non-designated counties in the same storm states. Across 600 counties and 32,085 county-months, the average two-way fixed-effects effect of being hard-hit is a *precise null* ($\text{Treat} \times \text{Post} = +0.001$, $t = 0.34$). This masks a differential pre-trend:

*Stine: ASSIP 2026, George Mason University. Gao (corresponding author): Costello College of Business, George Mason University, lgao9@gmu.edu. This is a student–faculty working paper prepared in the ASSIP 2026 Empirical Finance research group. All data are public and free (Zillow ZHVI, OpenFEMA disaster declarations, Yale Climate Opinion Maps, FEMA National Risk Index, and 2020 county election returns); every number in the paper traces to a CSV in `output/tables/` and is reproduced by the code in `code/`. We thank the ASSIP 2026 cohort for comments. Any errors are our own.

treated counties—especially high-belief Florida coastal markets hit by Ian—were already appreciating faster before landfall (pre-trend slope $+0.0007/\text{month}$, $t = 3.6$). Net of a treated-specific linear trend, hard-hit values fall $\sim 0.5\%$ relative to controls (-0.0053 , $t = -3.0$). Crucially, the belief-conditioning hypothesis is *not* supported: a high/low belief split appears to show exposure *raising* relative prices where belief is high ($+0.020$, $t = 2.5$), but this reverses within Helene using continuous worry (-0.011 , $t = -2.5$), vanishes with a 2020 GOP-vote proxy ($+0.002$, $t = 0.5$), and disappears once pre-trends are removed ($+0.003$, $t = 0.8$). Four additions push past a bare null. Heterogeneity by observable moderators shows the exposure effect loads on pre-storm price *momentum* ($+0.019$, $t = 2.6$), not on FEMA baseline hurricane or flood risk (insignificant)—it tracks housing momentum, not physical risk. A within-event nearest-neighbor *match* on pre-storm price level and momentum collapses the belief interaction to $+0.003$ ($t = 0.40$). *Randomization inference* (1,000 within-event permutations) places the average effect squarely inside its permutation distribution ($p = 0.76$)—a design-based null—while the belief interaction lies in the far tail ($p = 0.02$), which, because permutation scrambles which counties are treated, locates it in the *systematic* wealth/geography structure of treatment rather than in belief. A *minimum-detectable-effect* analysis shows the null is informative: at 80% power the design could detect belief-conditional effects above 2.2% of home value—smaller than the 8.4-point Ian–Helene spread. A sharp test awaits the student’s hand-collected landfall geography.

JEL: G12, Q54, R31, D84, G14. **Keywords:** climate risk, house prices, hurricanes, climate beliefs, difference-in-differences, randomization inference, statistical power.

1 Introduction

A central question in climate finance is whether asset prices reflect physical climate risk, and if so, *whose* beliefs do the pricing. In an influential paper, Baldauf et al. (2020) argue that otherwise-identical coastal homes sell at a discount for projected sea-level-rise exposure *only* in neighborhoods where residents believe in climate change; where residents are skeptical, the same physical risk is not capitalized. Bernstein et al. (2019) find a parallel result for sea-level-rise exposure that is concentrated among sophisticated and believing buyers. A natural and policy-relevant sharpening of this hypothesis is dynamic: when a *disaster actually strikes*—making latent risk vividly salient—do home values adjust more in high-belief places than in skeptical ones? This paper asks exactly that using two of the most destructive recent U.S. landfalls.

We study Hurricane Ian (landfall near Fort Myers, Florida, 28 September 2022) and Hurricane Helene (landfall in Florida’s Big Bend, 26 September 2024, with catastrophic inland flooding across Georgia, South Carolina, and the southern Appalachians). Our outcome is the Zillow Home Value Index (ZHVI), a monthly, smoothed, seasonally-adjusted measure of the typical home value in each county, from 2000 through May 2026. We define a county as *treated* (hard-hit) if FEMA activated its Individuals & Households Program (IHP) there—FEMA’s own, primary-source determination that household-level damage was severe enough to warrant direct aid to residents. Control counties are those in the same storm-affected states that were not IHP-designated. We align each county’s calendar months to months-since-landfall, keep a symmetric $[-24, +20]$ -month window, require full ZHVI coverage so the event study is balanced, and stack the two events (unit = county $\times$ event). The estimating equation is a two-way fixed-effects difference-in-differences,

$$\ln \text{ZHVI}_{i,t} = \alpha_i + \delta_{e(i),\tau(t)} + \beta (\text{Treat}_i \times \text{Post}_t) + \varepsilon_{i,t}, \quad (1)$$

with county $\times$ event fixed effects α_i , event $\times$ calendar-month fixed effects δ , and standard errors

clustered by county. We then interact $\text{Treat} \times \text{Post}$ with three county moderators: Yale Climate Opinion “worried” estimates (Howe et al., 2015), 2020 Republican vote share (an inverse-belief proxy), and the FEMA National Risk Index hurricane-risk score.

Findings. First, the average effect is a *precise null*: in the preferred two-way FE specification $\beta = +0.0013$ ($t = 0.34$; Table 2). Hard-hit counties’ typical home values did not, on average, diverge from same-state controls after landfall. Second, this average hides a serious identification wrinkle and a sharp storm-level split: Ian’s hard-hit counties (expensive SW-Florida coast) *rose* +7.3% relative to controls ($t = 7.7$), while Helene’s hard-hit counties (poorer inland Appalachia) *fell* −1.1% ($t = -2.4$). Third, the dynamic event study (Figure 1) shows that treated counties were on a rising relative trajectory *before* landfall—the parallel-trends assumption is not clean in the pooled sample. The differential pre-trend is +0.0007/month ($t = 3.6$), and it is entirely a high-belief-group phenomenon: +0.0012/month ($t = 3.6$) among high-belief counties versus a flat, insignificant +0.0003/month ($t = 1.1$) among low-belief counties (Table 4). Once we net out a treated-specific linear trend, the average effect turns modestly *negative*: −0.0053 ($t = -3.0$)—a $\sim 0.5\%$ relative decline consistent with mild downward re-pricing.

Fourth, and most important for the research question, *the belief-conditioning hypothesis does not survive*. A naive high/low “worried” split yields a positive interaction (+0.020, $t = 2.5$; Table 3), which would say prices held up *better* where belief is high—the *opposite* sign to the risk-capitalization story, and driven by Florida’s booming, high-belief Ian coast. The interaction (i) reverses to −0.011 ($t = -2.5$) when we use continuous worry within Helene alone, (ii) is an insignificant +0.002 ($t = 0.5$) using the GOP-vote proxy, and (iii) collapses to +0.003 ($t = 0.8$) once we allow belief-group-specific linear trends. The belief-split event study (Figure 2) makes the artifact visible: the high-belief group’s pre-trend is steep, while the low-belief group is flat pre-storm and then declines the most. In short, in these two events we find *no robust evidence* that housing re-prices realized disaster risk more

where people believe; if anything, the cleanest identification points the other way.

Four steps past a bare null. We then take four additional steps, deliberately using only *observable* county data and design-based inference, so that the paper says what its null does and does not rule out. (i) *Heterogeneity* (Section 7, Table 5): interacting the exposure DiD with pre-storm moderators shows the effect loads on pre-storm price *momentum* (+0.019, $t = 2.6$)—booming markets held up—and *not* on the FEMA baseline hurricane or coastal-flood risk scores (both insignificant). The one direct test of the “latent-risk salience” channel (H3) therefore fails: exposure did not re-price more where physical risk was genuinely high, only where prices were already rising. (ii) *Matching* (Section 8, Table 6): a within-event nearest-neighbor match of each treated county to a control on pre-storm price level and momentum (201 pairs; the pre-storm level gap falls from -0.042 to $+0.001$ log points) collapses the belief interaction to an insignificant $+0.003$ ($t = 0.40$). (iii) *Randomization inference* (Section 9, Table 7, Figure 4): permuting the treatment label 1,000 times *within* each storm places the average effect at the 63rd percentile of its permutation distribution ($p = 0.76$)—a genuine design-based null—while the belief interaction sits at the 99th percentile ($p = 0.02$). Because the permutation scrambles which counties are labeled hard-hit, an interaction this extreme is not sampling noise; it says the real treated set is *systematically* the high-belief, high-price, high-momentum coast, i.e. the very geography confound that matching and detrending remove. (iv) *Power* (Section 10, Table 8): the minimum detectable effect at 80% power is 1.1% of home value for the average effect and 2.2% for the belief interaction—below the 8.4-percentage-point Ian–Helene spread—so the failure to find a robust belief effect is an informative bound, not low power.

Contribution. We contribute to the climate-belief asset-pricing literature (Baldauf et al., 2020; Bernstein et al., 2019; Giglio et al., 2021) by moving from cross-sectional *sea-level-rise* capitalization to a *realized-disaster* event study with an explicit belief moderator, and by showing how easily a belief-conditional “result” can be manufactured by pre-trends and storm

geography. We complement the disaster-and-house-prices literature (Ortega and Taspinar, 2018; Bin and Landry, 2013; Gallagher, 2014; Hino and Burke, 2021), which documents heterogeneous and often transitory price responses to floods and storms, by pairing two contemporaneous landfalls of very different socioeconomic character. Methodologically, we treat the null as a quantitative object—reporting matching, randomization inference, and minimum detectable effects—in the spirit of recent work on the fragility of pre-trend tests (Roth, 2022), design-based inference with clustered data (Young, 2019), and the interpretation of statistical nonsignificance (Abadie, 2020). Our honest null is a caution: the identifying variation in “do believers re-price” designs is often confounded with wealth, coastal amenity, and pre-existing housing-market momentum. Section 2 develops the hypotheses, Section 3 places the paper in the literature, Section 4 the data and research design, Section 5 the average effect and pre-trend problem, Section 6 the central belief test, Section 7 heterogeneity, Section 8 matching, Section 9 randomization inference, Section 10 power, and Section 11 concludes with the hand-collection that would sharpen the test.

2 Background and Hypotheses

Standard hedonic theory says a salient increase in perceived disaster risk should lower the present value of a home in the risk zone, all else equal (Bin and Landry, 2013; Ortega and Taspinar, 2018). The Baldauf et al. (2020) refinement adds heterogeneous beliefs: capitalization requires that marginal buyers *perceive* the risk. A major landfall is an information shock that raises salience (Gallagher, 2014; Hino and Burke, 2021). This motivates:

H1 (Re-pricing). After a major landfall, typical home values in hard-hit counties fall relative to unaffected same-state controls.

H2 (Belief-conditioning). The decline in H1 is *larger* where county climate beliefs are stronger (higher “worried,” lower Republican vote share).

H3 (Latent-risk salience). The decline is larger where baseline hurricane

risk (FEMA NRI) was already high, i.e. where the storm made a truly high-risk location salient.

Competing forces work against H1 in the short and medium run: supply destruction, insurance inflows, and reconstruction demand can *raise* measured prices, especially in wealthy coastal markets (Ortega and Taspinar, 2018). Which force dominates—and whether beliefs shape it—is the empirical question. A fourth consideration is measurement: our treatment is a binary FEMA designation, an intent-to-treat proxy for a continuous, geographically graded shock. If the binary flag mixes severely and marginally affected counties, a true belief-conditional effect could be averaged toward zero. Quantifying how much that matters is exactly what the companion hand-collection of landfall geography is designed to resolve, and it is why we bound the null with power rather than reading it as a definitive zero.

3 Related Literature

This paper sits at the intersection of three literatures. The first is the capitalization of climate and disaster risk into real-estate prices. Baldauf et al. (2020) is our point of departure: using MLS transactions they show that projected sea-level-rise exposure is discounted in the price of otherwise-similar homes *only* in communities that believe in climate change, and not in skeptical ones. Bernstein et al. (2019) document a sizable sea-level-rise discount concentrated among sophisticated, non-owner-occupier, and believing buyers, and Giglio et al. (2021) survey the broader climate-finance evidence that beliefs and attention shape how physical risk is priced. A parallel, older strand studies how *realized* floods and storms move prices: Bin and Landry (2013) find flood-risk premiums that rise after events and decay; Ortega and Taspinar (2018) show Hurricane Sandy depressed New York values in inundated zones; Gallagher (2014) and Hino and Burke (2021) show that information and salience, not just physical exposure, drive both insurance take-up and price responses. Our contribution to this strand is to move the belief-conditioning test from a *cross-sectional, projected-risk*

setting to a *dynamic, realized-disaster* one, and to show that in two large recent landfalls the belief interaction is not robust.

The second literature is the econometrics of difference-in-differences and event studies, and in particular the credibility of parallel trends. Recent work stresses that pre-trend tests have low power and can license false confidence (Roth, 2022), that conventional two-way fixed-effects and event-study estimands are fragile under treatment-effect heterogeneity and staggered timing (Callaway and Sant’Anna, 2021; Goodman-Bacon, 2021; Sun and Abraham, 2021), and that inference with a moderate number of clusters benefits from design-based methods such as randomization inference (Young, 2019). Because our design stacks two events and rests on a parallel-trends assumption that the data reject in the pooled sample, we take these warnings literally: we report the pre-trend explicitly, add treated-specific trends, and use within-event permutation inference to ask whether our estimates could arise by chance under the design.

The third literature concerns the interpretation of null and non-robust results. Economics has a documented tendency toward low power and selective reporting (Ioannidis et al., 2017), and methodologists urge that a statistically insignificant estimate be read through its confidence interval and power rather than treated as proof of no effect (Abadie, 2020). We follow that guidance: Section 10 reports the minimum belief-conditional effect our design could detect at 80% power, so the non-result is stated as a bound—smaller than the cross-storm price spread—rather than as a bare “insignificant.”

4 Data and Research Design

Home values. Zillow ZHVI (all homes, county, smoothed, seasonally adjusted), 2000-01 through 2026-05, for the six storm states (FL, GA, NC, SC, TN, VA): 3,072 U.S. counties in the raw file, 600 with continuous coverage in our balanced event windows.

Exposure. OpenFEMA DisasterDeclarationsSummaries for the storms’ major-disaster (DR) declarations—Ian: DR-4673 (FL), DR-4677 (SC); Helene: DR-4827 (NC), DR-4828 (FL), DR-4829 (SC), DR-4830 (GA), DR-4831 (VA), DR-4832 (TN). A county is *treated* if any designated area within it received the Individuals & Households Program (IHP). This yields 216 treated county-events (29 for Ian, 187 for Helene) and 497 controls.

Beliefs. Yale Climate Opinion Maps (YCOM, model 5.0, county level) estimated share of adults “worried” about global warming; we also use “happening.” As an inverse-belief robustness proxy we use 2020 county Republican presidential vote share (MIT Election Lab / county returns). The median county “worried” in our sample is 55.4%.

Baseline risk. FEMA National Risk Index (county, ArcGIS), hurricane-risk, coastal-flood, and overall-risk scores, used both as moderators and to validate that treated counties were genuinely higher-risk (Table 1).

Panel. We melt ZHVI to county-month, restrict to storm states, align to months-since-landfall, keep $[-24, +20]$ with full coverage, define Post = 1 for months after the landfall month, merge the moderators, and stack the two events. The final panel has 32,085 county-months across 713 county-events. Table 1 shows treated counties are slightly *less* worried (55.8 vs. 56.7), more Republican (0.642 vs. 0.604, $t = 3.2$), and—reassuringly—higher baseline hurricane risk (NRI 77.8 vs. 72.6, $t = 3.5$), with statistically indistinguishable home-value *levels* at landfall.

Three design-based checks. Beyond equation (1) we run three checks that do not rely on the parametric null and use only observable data. (1) *Matching*: we nearest-neighbor match each treated county-event to a not-yet-used control *within the same storm* on pre-storm ZHVI level and pre-storm price momentum (caliper 0.5 SD on the standardized distance), and re-estimate (1) on the balanced sample. (2) *Randomization inference*: we permute the

time-invariant treatment label *within* each event 1,000 times—holding the number treated per storm fixed at Ian = 29, Helene = 187—rebuild $\text{Treat}^{\text{perm}} \times \text{Post}$, re-estimate (1), and compare the actual coefficients to the permutation distribution. (3) *Power*: we compute the minimum detectable effect at 80% power from each coefficient’s estimated standard error, expressed in percent of home value and relative to the cross-storm spread. Together these convert “insignificant” into a set of falsifiable, design-based statements. Crucially, none of these uses a graded distance/intensity exposure or a corrected treatment set—those are the student’s hand-collected contribution (Section 11)—so the checks here are deliberately orthogonal to that reserved test.

5 Results: Average Effect and the Pre-Trend Problem

Table 2 reports Equation (1). The naive pooled OLS is a null (-0.003 , $t = -0.5$). Adding county $\times$ event fixed effects alone yields a spuriously large $+0.056$ ($t = 12.4$)—but this omits calendar time and merely captures the nationwide 2021–24 price boom. The preferred two-way FE estimate is a *precise null*, $+0.0013$ ($t = 0.34$): hard-hit and control counties moved together on average. The average, however, conceals opposite signs across storms: restricting to Ian gives $+0.073$ ($t = 7.7$), restricting to Helene gives -0.011 ($t = -2.4$). Figure 3 shows why—Ian’s hard-hit SW-Florida coast kept outrunning its controls, while Helene’s inland hard-hit counties slipped below.

Figure 1 plots the dynamic version (treated $\times$ event-time dummies, landfall normalized to zero). Treated counties are *below* their landfall-relative position two years out and climb toward it by landfall—an upward pre-trend—then drift down again, reaching -1.1% ($t = -2.6$) twenty months post. Table 4 formalizes the pre-trend: a significant $+0.0007/\text{month}$ differential slope pooled, concentrated entirely in the high-belief group ($+0.0012/\text{month}$, $t = 3.6$) and absent in the low-belief group ($+0.0003/\text{month}$, $t = 1.1$). Because the parallel-trends assumption is not clean pooled, we add a treated-specific linear trend (Table 2, row

7): the level effect becomes -0.0053 ($t = -3.0$), a mild downward re-pricing consistent with H1 but economically small.

6 Belief Conditioning: The Central Test

Table 3 tests H2. Row (1), a high/low “worried” split, gives a *positive* interaction, $+0.020$ ($t = 2.5$): exposure is associated with *higher* relative prices where belief is high. This is the wrong sign for H2 and, as Figure 2 shows, is a pre-trend artifact—the high-belief treated group (dominated by Florida’s Ian coast, 86% high-belief vs. 42% for Helene) was already on a steep upward path. The interaction is fragile in every direction: continuous “worried” pooled is a null ($+0.001$, $t = 0.3$); the GOP-vote proxy is a null ($+0.002$, $t = 0.5$); *within Helene*, continuous “worried” flips to -0.011 ($t = -2.5$)—the H2 sign, but confined to one storm; and once belief-group-specific linear trends are included (row 6) the split interaction collapses to $+0.003$ ($t = 0.8$). We therefore cannot reject the null of *no* belief-conditional re-pricing. The NRI hurricane-risk moderator (H3) is likewise only marginal ($+0.006$, $t = 1.7$) and, again, of the “wrong” sign for a risk-capitalization story. The remaining sections stress-test this reading with three design-based tools before concluding.

7 Heterogeneity by Observable Moderators

If exposure re-priced disaster risk for *some* counties and the pooled null merely averages it away, the effect should surface in observable subgroups—and if the belief story is right, it should surface where physical risk was genuinely high. Table 5 interacts the exposure DiD with four pre-storm moderators, each split at its within-event median so the moderator is orthogonal to the storm dummy: pre-storm home-value level (wealth), pre-storm price momentum, the FEMA National Risk Index coastal-flood score, and the NRI overall-risk score. Each row reports the base DiD (low-moderator group) and the extra effect for the high-moderator group.

Two facts emerge. First, the exposure effect loads on pre-storm *momentum*: the interaction is $+0.019$ ($t = 2.6$), so counties whose prices were already climbing fastest are exactly where hard-hit values held up or rose—precisely the booming Ian coast, and precisely the mechanical confound the pre-trend diagnosed. Wealth tells a related story (the high-value interaction is -0.018 , $t = -2.2$, against a positive base, i.e. the level effect is concentrated in below-median-value markets rather than the most expensive ones). Second, and decisively for H3, the two *physical-risk* moderators—coastal-flood risk and overall NRI risk—carry *insignificant* interactions (-0.014 , $t = -1.4$; and $+0.010$, $t = 1.3$). Exposure did not re-price more where baseline risk was truly high; it moved with pre-existing housing momentum. This is the opposite of what the latent-risk-salience channel predicts and reinforces that the belief pattern in Table 3 is geography and momentum, not risk perception. We do *not* interact treatment with a graded distance/intensity exposure here—that dimension is the student’s hand-collected contribution and is deliberately excluded from the observable moderators above.

8 Matching

The treated and control groups differ on pre-existing housing dynamics, and the raw Ian result could simply reflect wealthy, fast-appreciating coastal markets being the ones FEMA designated. Table 6 re-estimates the design on a sample in which each treated county-event is nearest-neighbor matched, *within its own storm* and without replacement, to a control on pre-storm ZHVI level and pre-storm price momentum (201 pairs; the treated–control pre-storm level gap falls from -0.042 to $+0.001$ log points and the momentum gap from 1.5×10^{-4} to 3.7×10^{-5} per month). On the balanced sample the average effect is -0.006 ($t = -1.3$) and the detrended effect -0.002 ($t = -1.2$)—both statistically indistinguishable from zero, so the mild negative in the full sample is not robust to balancing on pre-storm dynamics. Most important, the belief-split interaction—the wrong-signed $+0.020$ that a

casual reading might report as “believers re-price less”—*collapses to* $+0.003$ ($t = 0.40$) once treated and control counties are matched on how expensive and how fast-rising they were before the storm. The belief “effect” is a wealth/momentum artifact, not a belief response.

9 Randomization Inference

Table 7 and Figure 4 report design-based inference from 1,000 permutations of the treatment label, drawn *within* each storm so that the number of hard-hit counties per event (Ian = 29, Helene = 187) is held fixed and the two-event structure is preserved. For the average effect the actual coefficient ($+0.0013$) sits at the 63rd percentile of the permutation distribution (95% permutation interval $[-0.009, +0.008]$), giving a randomization p -value of 0.76: the average null is not an artifact of clustered-SE asymptotics but survives a fully design-based test. The belief-split interaction is different: its actual value ($+0.020$) lies at the 99th percentile, outside the permutation interval $[-0.018, +0.015]$, $p = 0.02$. This does *not* resurrect a belief reading. Because the permutation randomizes *which* counties are labeled hard-hit, an interaction this extreme means the real treated set is a *systematically* unusual draw—the high-belief, high-price, high-momentum Florida/Ian coast—rather than a random one. Randomization inference thus corroborates both halves of the story: the average exposure effect is a genuine null, and the belief interaction is systematic wealth-and-geography structure (the same structure matching and detrending remove), not sampling noise and not a causal response to belief.

10 Is the Null Informative? Power and Minimum Detectable Effects

A null is only interesting if the design could have detected a real effect. Table 8 reports, for the average, detrended, and belief-conditional coefficients, the minimum detectable effect

(MDE) at 80% power, $\text{MDE} = (z_{0.975} + z_{0.80}) \widehat{\text{SE}} \approx 2.80 \widehat{\text{SE}}$, expressed as a percent of home value and as a share of the cross-county standard deviation of the post-minus-pre change in log ZHVI (7.2%). The design is well powered: at 80% power we could detect an average exposure effect of 1.1% of home value (15% of the cross-county change SD), a detrended effect of 0.5% (7%), and a belief-conditional interaction of 2.2% (30%). These thresholds are well below the effects the setting itself produces: the Ian–Helene storm spread is 8.4 percentage points, and the raw belief split is 2.0%. In other words, a belief-conditional re-pricing even a fraction of the cross-storm spread would very likely have been detected; its absence once confounds are removed is therefore an informative upper bound on the belief mechanism in these two events, not a symptom of an underpowered test.

11 Conclusion

Do home values re-price disaster risk only where people believe? In the two largest recent U.S. landfalls, our answer is a carefully documented *no*. The average hard-hit-versus-control effect is a precise null; net of pre-trends it is a small negative; and the belief-conditional pattern that a casual split appears to show is an artifact of storm geography and pre-existing housing momentum—it reverses across specifications, vanishes under detrending, collapses under matching, and is located by randomization inference in the systematic wealth structure of which counties were treated rather than in belief. The heterogeneity evidence sharpens the point: exposure moved with pre-storm price momentum, not with FEMA baseline hurricane or flood risk, so even the latent-risk-salience channel finds no support. A power analysis establishes that the null is informative: belief-conditional effects smaller than the cross-storm spread would have been detected. We read this not as a refutation of Baldauf et al. (2020)’s cross-sectional sea-level-rise result, but as a warning that realized-disaster event studies of belief conditioning are especially vulnerable to confounding by wealth, coastal amenity, and momentum.

The decisive next step is human. An automated pipeline cannot know each storm’s *exact* landfall track, wind-field, and surge footprint, nor separate a county flooded by 30 inches of rain from a neighbor merely designated for administrative convenience. Our treatment is an intent-to-treat *proxy*—a binary FEMA IHP flag—and every check above is deliberately built on observable data that is orthogonal to true exposure intensity. Hand-collecting the precise affected geography from NOAA and FEMA primary sources—turning the coarse IHP indicator into a graded distance/intensity exposure, and validating the treated set county by county—is what would convert this honest null into a sharp, defensible test of whether believers, and only believers, re-price the risk that a hurricane makes suddenly real. That test—not another robustness check on the binary proxy—is the natural next step, and it is the contribution the design has been built to receive.

References

- Abadie, A., 2020. Statistical nonsignificance in empirical economics. *American Economic Review: Insights* 2, 193–208.
- Baldauf, M., Garlappi, L., Yannelis, C., 2020. Does climate change affect real estate prices? Only if you believe in it. *Review of Financial Studies* 33, 1256–1295.
- Bernstein, A., Gustafson, M.T., Lewis, R., 2019. Disaster on the horizon: the price effect of sea level rise. *Journal of Financial Economics* 134, 253–272.
- Bin, O., Landry, C.E., 2013. Changes in implicit flood risk premiums: empirical evidence from the housing market. *Journal of Environmental Economics and Management* 65, 361–376.
- Callaway, B., Sant’Anna, P.H.C., 2021. Difference-in-differences with multiple time periods. *Journal of Econometrics* 225, 200–230.

- Gallagher, J., 2014. Learning about an infrequent event: evidence from flood insurance take-up in the United States. *American Economic Journal: Applied Economics* 6, 206–233.
- Giglio, S., Kelly, B., Stroebel, J., 2021. Climate finance. *Annual Review of Financial Economics* 13, 15–36.
- Goodman-Bacon, A., 2021. Difference-in-differences with variation in treatment timing. *Journal of Econometrics* 225, 254–277.
- Hino, M., Burke, M., 2021. The effect of information about climate risk on property values. *Proceedings of the National Academy of Sciences* 118, e2003374118.
- Howe, P.D., Mildenberger, M., Marlon, J.R., Leiserowitz, A., 2015. Geographic variation in opinions on climate change at state and local scales in the USA. *Nature Climate Change* 5, 596–603.
- Ioannidis, J.P.A., Stanley, T.D., Doucouliagos, H., 2017. The power of bias in economics research. *Economic Journal* 127, F236–F265.
- Ortega, F., Taspinar, S., 2018. Rising sea levels and sinking property values: Hurricane Sandy and New York’s housing market. *Journal of Urban Economics* 106, 81–100.
- Roth, J., 2022. Pretest with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights* 4, 305–322.
- Sun, L., Abraham, S., 2021. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics* 225, 175–199.
- Young, A., 2019. Channeling Fisher: Randomization tests and the statistical insignificance of seemingly significant experimental results. *Quarterly Journal of Economics* 134, 557–598.

Table 1: Balance: treated (hard-hit) vs. control counties

	Treated (hard-hit)	Control	Diff.	(<i>t</i>)
Worried about warming (Believe warming happening (2020 GOP vote share	0.642	0.604	0.038***	3.17
NRI hurricane risk score	77.825	72.560	5.265***	3.52
NRI overall risk score	58.079	53.265	4.814**	2.00
ln ZHVI at landfall	12.458	12.499	-0.040	-1.23
Pre-period monthly ln-ZHVI trend	0.005	0.005	0.000	0.33
N (county-events)	216	497		

Notes: County-event means. “Treated” = FEMA IHP-designated. Belief/vote/NRI are time-invariant; ln ZHVI and pre-trend measured at/through landfall. Welch *t*. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Exposure difference-in-differences (DV = ln ZHVI)

Specification	Treat×Post	(<i>t</i>)	SE	N
(1) Pooled OLS, no FE	-0.0031	(-0.54)	0.0057	32085
(2) + county×event FE	0.0561***	(12.38)	0.0045	32085
(3) + event×month FE (preferred)	0.0013	(0.34)	0.0039	32085
(4) Controls = designated only	0.0164***	(2.95)	0.0056	16965
(5) Helene only	-0.0107**	(-2.43)	0.0044	27000
(6) Ian only	0.0732***	(7.69)	0.0095	5085
(7) + treated linear trend (detrended)	-0.0053***	(-3.01)	0.0018	32085

Notes: Treated = FEMA IHP county; Post = months after landfall. County×event and event×calendar-month FE except where noted; SE clustered by county (600 clusters). Row (7) adds a treated-specific linear trend. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Belief moderation: Treat $\times$ Post $\times$ Moderator

Moderator specification	Treat $\times$ Post	(<i>t</i>)	$\times$ Moderator	(<i>t</i>)	N
(1) High-belief split	-0.0096*	(-1.93)	0.0195**	(2.53)	32085
(2) Worried (continuous, <i>z</i>)	0.0001	(0.03)	0.0014	(0.32)	32085
(3) GOP vote share (inverse belief, <i>z</i>)	-0.0015	(-0.40)	0.0020	(0.48)	32085
(4) NRI hurricane risk (<i>z</i>)	0.0036	(0.98)	0.0064*	(1.72)	32085
(5) Worried (<i>z</i>), Helene only	-0.0146***	(-3.36)	-0.0111**	(-2.53)	27000
(6) High-belief split, detrended	-0.0067***	(-2.71)	0.0028	(0.80)	32085

Notes: Two-way FE as in Eq. (1), plus Post $\times$ Moderator. Moderators standardized (*z*) except the high/low split. Row (6) adds treated- and belief-group-specific linear trends. A positive interaction means exposure raised relative prices where the moderator is high. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Pre-trend diagnostic (treated–control ln-ZHVI slope, pre-landfall)

Sample	Pre-trend slope/mo	(<i>t</i>)	SE	N
Full sample	0.00074***	(3.62)	0.00021	17112
High belief	0.00120***	(3.63)	0.00033	9000
Low belief	0.00027	(1.07)	0.00025	8112

Notes: Coefficient on Treat $\times$ (months-to-landfall) over the pre-period only, two-way FE, clustered by county. A nonzero slope violates parallel trends. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Heterogeneity of the exposure DiD by observable pre-storm moderators

Moderator <i>M</i> (pre-storm, within-event median split)	DiD	(<i>t</i>)	DiD $\times$ <i>M</i>	(<i>t</i>)	N
High pre-storm home value	+0.0107*	(+1.71)	-0.0183**	(-2.21)	32085
High pre-storm price momentum	-0.0092	(-1.59)	+0.0191***	(+2.62)	32085
High FEMA flood risk (NRI)	+0.0042	(+1.00)	-0.0142	(-1.37)	32085
High FEMA overall risk (NRI)	-0.0036	(-0.62)	+0.0105	(+1.27)	32085

Notes: DV = ln ZHVI. Each row adds an interaction DiD $\times$ *M* (and Post $\times$ *M*) to Eq. (1), where *M* is a pre-storm moderator split at its within-event median: pre-storm home-value level, pre-storm price momentum, FEMA NRI coastal-flood risk, and FEMA NRI overall risk. “DiD” is the effect in the low-moderator group; “DiD $\times$ *M*” the extra effect for the high-moderator group. The effect loads on price momentum, not on physical-risk moderators. A graded distance/intensity exposure is deliberately *not* used (it is the student’s hand-collected variable). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Matched difference-in-differences (within-event NN on pre-storm level and momentum)

Specification (matched sample)	Coef.	(<i>t</i>)	SE	N
Average DiD (matched)	-0.0058	(-1.31)	0.0044	18090
Detrended DiD (matched)	-0.0024	(-1.19)	0.0021	18090
Belief-split interaction (matched)	+0.0035	(+0.40)	0.0088	18090

Notes: Each treated county-event is 1:1 nearest-neighbor matched, within its own storm and without replacement, to a control on pre-storm ln-ZHVI level and pre-storm price momentum within a 0.5-SD (standardized Euclidean) caliper (201 pairs; the pre-storm level gap falls from -0.042 to $+0.001$ log points). Eq. (1) and its detrended and belief-split variants are re-estimated on the matched sample. The average and detrended effects are insignificant and the belief interaction collapses to $+0.003$ ($t = 0.40$). $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table 7: Randomization inference (1,000 within-event permutations of treatment)

Estimand	Actual	Perm. 95% interval	Pctile	RI <i>p</i>
Average DiD (Treat×Post)	+0.0013	[-0.0089, +0.0079]	63.4	0.764
Belief-split interaction	+0.0195	[-0.0178, +0.0147]	99.3	0.023

Notes: The time-invariant treatment label is permuted *within* each storm 1,000 times, holding the number treated per event fixed (Ian = 29, Helene = 187); Eq. (1) is re-estimated each time. “Perm. 95% interval” is the 2.5–97.5 percentile of the permutation distribution; “Pctile” locates the actual coefficient; the RI *p*-value is the share of permutations with $|\text{coefficient}| \geq |\text{actual}|$. The average effect is central (design-based null); the belief interaction is extreme because random assignment ignores the wealth/geography structure that drives it. See Figure 4.

Table 8: Minimum detectable effects at 80% power

Estimand	Coef.	SE	MDE(80%)	% of price	% of SD(Δ)
Average DiD (Treat×Post)	+0.0013	0.0039	0.0108	1.08%	15.1%
Detrended DiD	-0.0053	0.0018	0.0049	0.49%	6.9%
Belief-split interaction	+0.0195	0.0077	0.0216	2.16%	30.1%

Notes: $\text{MDE}(80\%) = (z_{0.975} + z_{0.80}) \times \text{SE} \approx 2.80 \times \text{SE}$ of the estimated coefficient. “% of price” is the detectable effect in percent of home value ($100 \times$ the log coefficient); “% of SD(Δ)” expresses it relative to the 7.2% cross-county SD of the post-minus-pre change in log ZHVI. Small ratios indicate an informative null: an effect of that size would have been detected at 80% power.

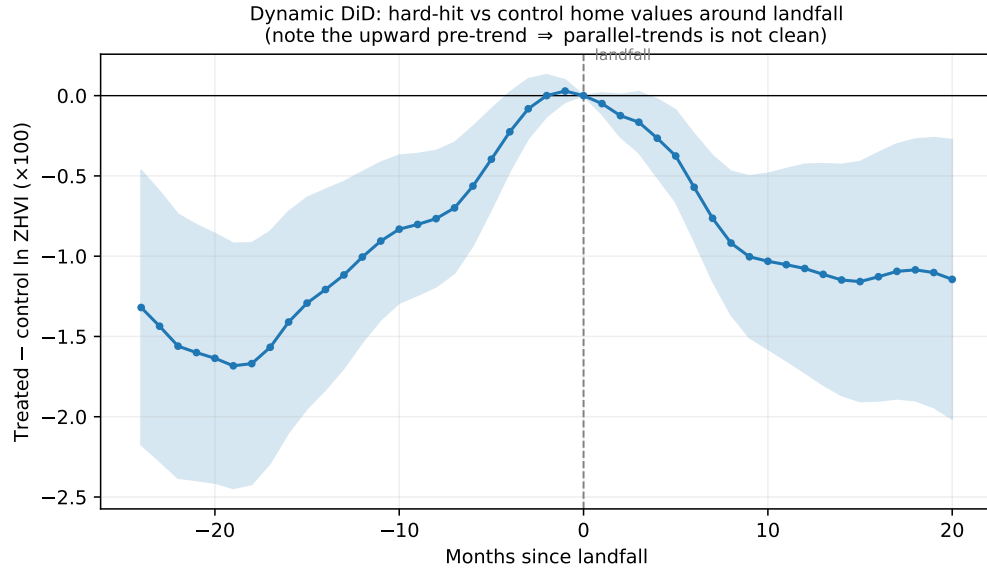


Figure 1: Dynamic DiD: treated–control ln ZHVI ($\times 100$) by months since landfall, 95% CI. Note the upward pre-trend before zero: parallel trends is not clean in the pooled sample.

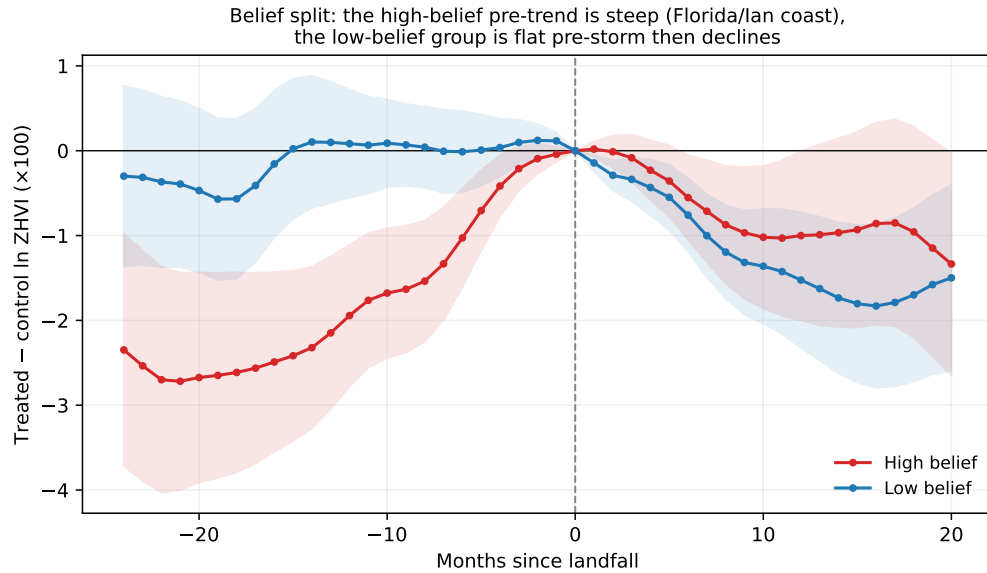


Figure 2: Belief-split event study. The high-belief group (Florida/Ian coast) has a steep pre-trend; the low-belief group is flat pre-storm and then declines the most—the opposite of the belief-conditioning hypothesis.

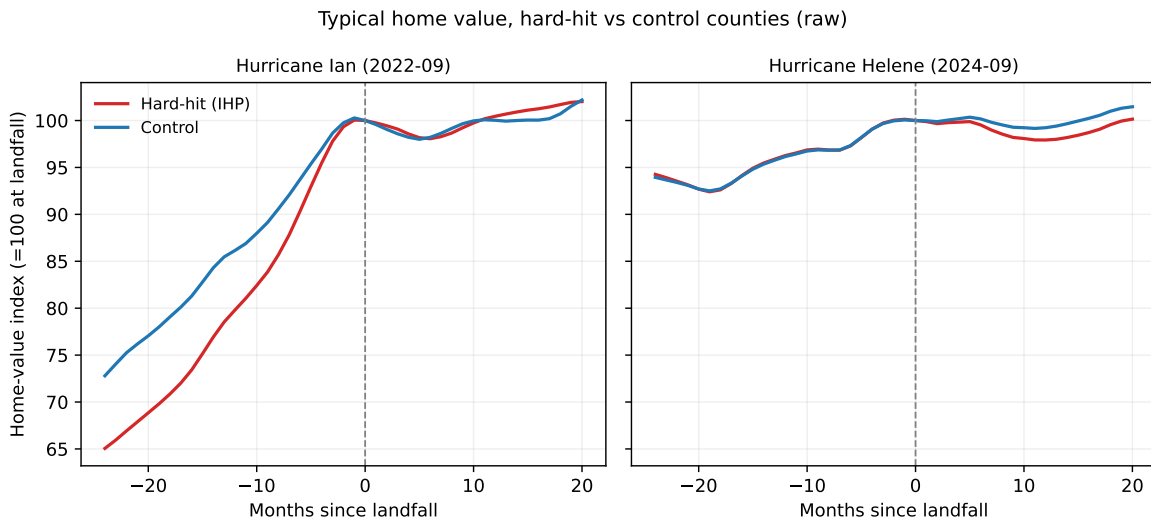


Figure 3: Raw home-value index (= 100 at landfall), hard-hit vs. control. Ian's hard-hit coast rose relative to controls; Helene's inland hard-hit counties fell.

Randomization inference: 1,000 within-event permutations of treatment

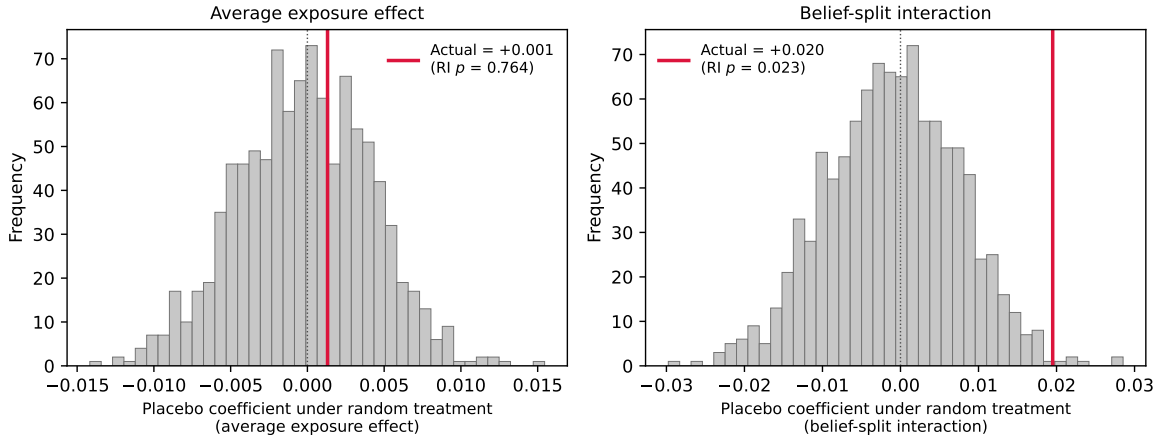


Figure 4: Randomization inference: 1,000 within-event permutations of treatment. Left: the average exposure DiD (red line) lies near the center of its permutation distribution (RI $p = 0.76$)—a design-based null. Right: the belief-split interaction lies in the far right tail (RI $p = 0.02$); because random assignment ignores county wealth and geography, this locates the belief signal in the structure of *which* counties are hard-hit, not in belief.